

ADITYA SHARMA

AI/ML Engineer | Computer Vision Enthusiast | Python Developer

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ABOUT ME

Computer Science undergraduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, Python Development, TensorFlow, PyTorch, and Full-Stack Application Development. Currently working as an AI/ML Intern developing image processing and computer vision solutions. Passionate about building scalable AI-powered products and solving real-world problems through data-driven technologies.

SKILLS

Languages: Python, C++, Java, SQL

Backend: FastAPI, Flask, Node.js, REST APIs

AI/ML: TensorFlow, PyTorch, Scikit-learn, OpenCV, CNNs, Deep Learning, Computer Vision

Data Science: Pandas, NumPy, Matplotlib, Seaborn

Databases: MySQL, SQLite

Tools: Git, GitHub, Android Studio, VS Code, Jupyter Notebook

EXPERIENCE

AI/ML Intern, AiAssured TestAlng

May 2026 – Present

- Developing and evaluating computer vision and image processing models for real-world AI applications, improving baseline accuracy by 12%.
- Working with Python, OpenCV, TensorFlow, and PyTorch to build and optimize machine learning pipelines, reducing inference time by 20%.
- Performing data preprocessing, model training, testing, and performance analysis on image datasets exceeding 10,000 samples.
- Contributing to AI-driven solutions focused on image classification, object detection, and visual intelligence systems.

PROJECTS

Dr Drive AI-Powered Vehicle Inspection & Predictive Maintenance Platform

March 2026 – Present

- Developed a full-stack **Android application** to analyze vehicle health using On-Board Diagnostics (OBD) data and computer vision.
- Implemented AI models to detect mechanical faults and predict failures with 85% accuracy, estimating repair costs based on real-time data.
- Integrated image-based damage detection for dents, scratches, and structural issues using deep learning techniques.
- Built a smart valuation system utilizing a dataset of 50,000+ vehicle records to estimate fair resale price and detect potential listing fraud.

Traffic Signal Waiting Time Analyzer

Feb 2026 – March 2026

- Engineered a C++ traffic simulation engine utilizing Queue data structures to model multi-lane intersections and track vehicle arrival times.
- Implemented and benchmarked scheduling algorithms (Fixed-Time, Greedy, and Priority Queue), applying Prefix Sum arrays to efficiently compute cumulative wait times.
- Developed a robust data pipeline connecting the C++ backend to a Python (Streamlit) frontend, parsing JSON outputs for real-time monitoring.

AI Powered Disaster Management Robot

Sept 2025 – Nov 2025

- Designed an AI-based system to detect and locate people trapped under debris with a 92% success rate in simulated disaster scenarios.
- Trained ML models to accurately monitor **pulse, heartbeat, oxygen, and Carbon Dioxide (CO2) levels** for real-time survivor detection.
- Integrated computer vision techniques for identifying human presence using advanced image detection algorithms in low-light environments.
- Focused on building a scalable and life-saving hardware-software solution for emergency response and search-and-rescue operations.

CERTIFICATIONS

Certificate of Finalist Statistella Data Analytics Competition (IIT BHU)

Jan 2026

Unstop

Advanced Data Analytics

Jul 2025

Google

Google AI Essentials

Dec 2024

Google

EDUCATION

Jaypee Institute of Information Technology

2024 – 2029

B.Tech + Integrated M.Tech in Computer Science

CGPA: 7.56

Masai School IIT Patna

Expected: 2026

Applied AI & ML Essentials

Delhi Public School Ghaziabad

2023

Senior Secondary (Class XII) (CBSE)

Percentage: 92.2%